Poll Everywhere Quizzes

At the beginning of each lecture:

1. Use your device to log in to PollEverywhere (PE). Enter your ANU email (uXXXXXXX@anu.edu.au); PE supports ANU SSO.

   https://pollev.com/anuiml2026



Photographing or recording the quiz questions is not permitted

Generalisation, VC dimension and PAC learning

Lecture 3 - Quiz 1 (120 seconds)

L3Q1 - A laboratory screens 5 000 candidate genetic markers on a sample of patients and reports the one that predicts the disease best in that sample. A second laboratory had committed in advance to a **single** marker, suggested by earlier biological work, and tests it on the **same** patients, obtaining an equally good prediction. Both report the same in-sample error. Which report is entitled to a stronger probabilistic guarantee?

- A) Both equally: every one of the 5 000 markers was written down before the records were examined, so the requirement that a hypothesis be fixed in advance is met by the one reported.
- B) The first: screening 5 000 markers searches a far larger part of the hypothesis set, so the marker reported is much closer to the best one available, and such a wide search supports a stronger conclusion.
- C) The second: the first reports the smallest in-sample error among 5 000 hypotheses evaluated on the same sample, so it was chosen *using* the data, and its guarantee is weakened by a union bound over all 5 000 candidates.
- D) Neither: a guarantee requires the hypothesis to be fixed before the data is *collected*, and both laboratories worked on patients whose records already existed, so neither report is covered.

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Lecture 3 - Quiz 2 (120 seconds)

L3Q2 - A model has finite VC dimension, hence is PAC learnable. A team collects a sample of the size prescribed by the theory, runs ERM, and the rule it returns makes **many errors, in sample and out of sample alike**. Which statement is consistent with the theory?

- A) The guarantee has been violated: reaching the prescribed sample size is precisely what drives the in-sample error to zero, so a large in-sample error means that one of the assumptions has failed, most plausibly that the sample is i.i.d.
- B) The sample size prescribed by the theory depends on the distribution generating the data: it was computed here for the wrong one, and for the distribution actually at work it would have demanded far more examples.
- C) Nothing is inconsistent: finite VC dimension makes the two errors close to each other, which they are, and the theory then says that both of them will come down together once a larger sample is collected.
- D) Nothing is inconsistent: the guarantee is only that the rule is nearly as good as the best rule *in the model*, and if the model is a poor description of the phenomenon, that best rule already makes many errors, which no amount of data repairs.

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Bias, variance, model selection and error estimation

Lecture 3 - Quiz 3 (120 seconds)

L3Q3 - A team fits a very flexible model. Each time the experiment is repeated on a fresh sample, the training fit is **perfect**, but the rule that is learned looks completely different. What is the right assessment?

- A) The instability is a property of the sample size alone and not of the hypothesis set: the only legitimate response is to collect more data, since restricting the hypothesis set would introduce a bias, a price that is never worth paying.
- B) It is a large estimation error (variance): a smaller hypothesis set does reduce it, but an arbitrary one may sit far from the target and have a large bias instead; a remedy is to choose the set based on prior information, or via model selection.
- C) The model is too flexible and must be restricted; among the smaller candidate sets, the one to keep is the one whose fitted hypothesis attains the smallest in-sample error, since that is what measures how well a model describes the data.
- D) A perfect fit on every sample shows that the hypothesis set can fit any labelling whatsoever, so its VC dimension is infinite and nothing can be learned about this phenomenon, whatever the sample size.

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Lecture 3 - Quiz 4 (120 seconds)

L3Q4 - A team splits its data into training, validation and test. It selects a model using the validation set, refits, and measures $E_{\text{test}} = 0.15$. Disappointed, it tries three further models, each trained in the same way and evaluated on the **same** test set, and reports the best of the four, $E_{\text{test}} = 0.09$. Is the reported 0.09 an honest estimate of E_{out}?

- A) Yes: no model saw the test labels while it was being trained, and all that is required for the estimate to be honest is that the hypothesis be fixed before the test data is used, which holds for each of the four separately.
- B) Yes: each of the four test errors is an unbiased estimate of the out-of-sample error of its own rule, so reporting the smallest of four unbiased estimates is simply reporting the most accurate figure the team has obtained.
- C) No: with respect to the test sample the reported rule is not a fixed hypothesis but the winner of a comparison made on that sample, so 0.09 is optimistically biased. The test set has been spent as a second validation set, and an honest figure requires data untouched by all four decisions.
- D) No, but the remedy is simple: report the average of the four test errors instead of the smallest of them, since averaging cancels the advantage that the winner obtained by having been selected.

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